Should You Switch To INTEL's Latest EMBEDDED BOARDS?

Embedded applications with smaller footprints and reduced power requirements, such as SMARC, Qseven, COM Express Compact/Mini, and Pico-ITX SBCs, get a boost with the launch of Intel's latest batch of Elkhart Lake processors. Replacing the previous Apollo Lake generation, they not only provide increased processing performance with decreased power requirements for industrial IoT applications, but also enable true real-time operation by supporting real-time connectivity and real-time hypervisor technology

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Delivering an impressive increase in performance, while requiring much less power, Intel's Elkhart Lake based trio of low power processors—Atom, Celeron, and Pentium—accelerate multithreaded workload processing by +50% compared to the previous generation, due to implementation of new microarchitecture on four cores. For single-thread applications, the high clock rates of up to 3GHz are advantageous and help increase performance by up to +70%. This elevated processing performance can be leveraged in very cold and hot environments, with trouble-free operation from -40 to +85°C, which

TABLE 1
INTEL ATOM x6000 VS E3900

	Intel Atom x6000 Series		Intel Atom E3900
	LPDDR4 / 4x 4267 MT/s	DDR4 3200 MT/S	DDR3L ECC 1600 MT/S
100% read (cache miss = 0%)	41 GB/s	30 GB/s	14 GB/s
100% write (cache miss = 0%)	35 GB/s	26 GB/s	16 GB/s
66/33 read/write (cache miss = 0%)	31 GB/s	23 GB/s	15 GB/s

TABLE 2
NEW INTEL ATOM, CELERON, AND PENTIUM PROCESSORS

Processor	Cores/ Threads	Clock [GHz] (Base/Boost)	CPU L2 Cache (MB)	GFE Execution Units	TDP (W)
Intel Atom X6425E	4	1.8 / 3.0	1.5	32	12
Intel Atom X6413E	4	1.5 /3.0	1.5	16	9
Intel Atom X6211E	2	1.2 /3.0	1.5	16	6
Intel Atom X6425RE	4	1.9/ -	1.5	32	12
Intel Atom X6414RE	4	1.5/ -	1.5	16	9
Intel Atom X6212RE	2	1.2 / -	1.5	16	6
Intel Pentium N6415	4	1.2 / 3.0	1.5	16	6.5
Intel Celeron N6211	2	1.2 / 3.0	1.5	16	6.5
Intel Pentium J6425	4	1.8 / 3.0	1.5	32	10
Intel Celeron J6413	4	1.8/ 3.0	1.5	16	10